Independent Study Addresses Causes of Challenges in Asia Pacific Flagship CCS Project

The Gorgon **carbon capture and storage** (CCS) project has pioneered the large-scale implementation of CCS in the Asia Pacific region. The late start of its CO_2-injection program, injection rates below target, and complexity in treatment of produced water from the regional aquifer, however, have contributed to a higher cost and a degree of media skepticism. In the complete paper, the authors provide an overview of some of the causes behind the challenges encountered to date, highlighting that these issues primarily relate to risk and pressure management of the recipient reservoir of the **produced water** from the Dupuy formation rather than from the technologies used for capture, transport, injection, and storage.

Introduction

Thirty to 40 operational sites worldwide use CCS technology primarily focused on precombustion projects, including natural gas processing, but the technology has been successfully deployed on post-combustion projects. The International Energy Agency's Sustainable Development Scenario suggests that a capacity of 6.2 Gtpa will be required for these projects by 2050 to meet the net-zero target. Globally, the current installed capacity is approximately 44 mtpa, which must increase to 6.2 Gtpa with an estimated investment of $4.5 trillion by 2030. The current number of CCS projects at the time of writing is between 40 and 45 worldwide, an increase from 24 in 2017.

Europe and North America have been pioneers in CCS and net-zero pursuit and have supported the projects through regulations and incentives. Australia has led its Asian peers and, with adoption and ratification of the London Protocol in 2009 and the establishment of Australian Carbon Credit Units, has begun the CCS journey earlier. Australia also is becoming more attractive as a commercial CO_2-storage destination for Asian countries, specifically Japan and South Korea. Table 1 of the complete paper shows projects under study, sanctioned, or operational in Southeast Asia and Australia as of 2024, and **Fig. 1** of this synopsis locates these projects.

CCS Storage Assessment and Challenges

Determining scale and feasibility is important for any project to establish CCS storage capacity and evaluation. Geological characterization also is required to assess containment and storage capacity and efficiency. Measuring and monitoring of CCS is another crucial part of project planning, design, and execution, because regulatory requirements and any risk of leakage or loss of containment must be proactively addressed.

This article, written by JPT *Technology Editor Chris Carpenter, contains highlights of paper* ***SPE 225882****, "Gorgon CCS Project: An Independent View of Why the Asia Pacific Flagship CCS Project Is Not a Carbon-Storage Issue," by* ***Ali Sabzabadi,*** *SPE,* ***Ian Gladman,*** *and* ***Bill Billingsley,*** *RISC Advisory. The paper has not been peer reviewed.*

The complete paper is free to SPE members until 31 January 2026 at jpt.spe.org.

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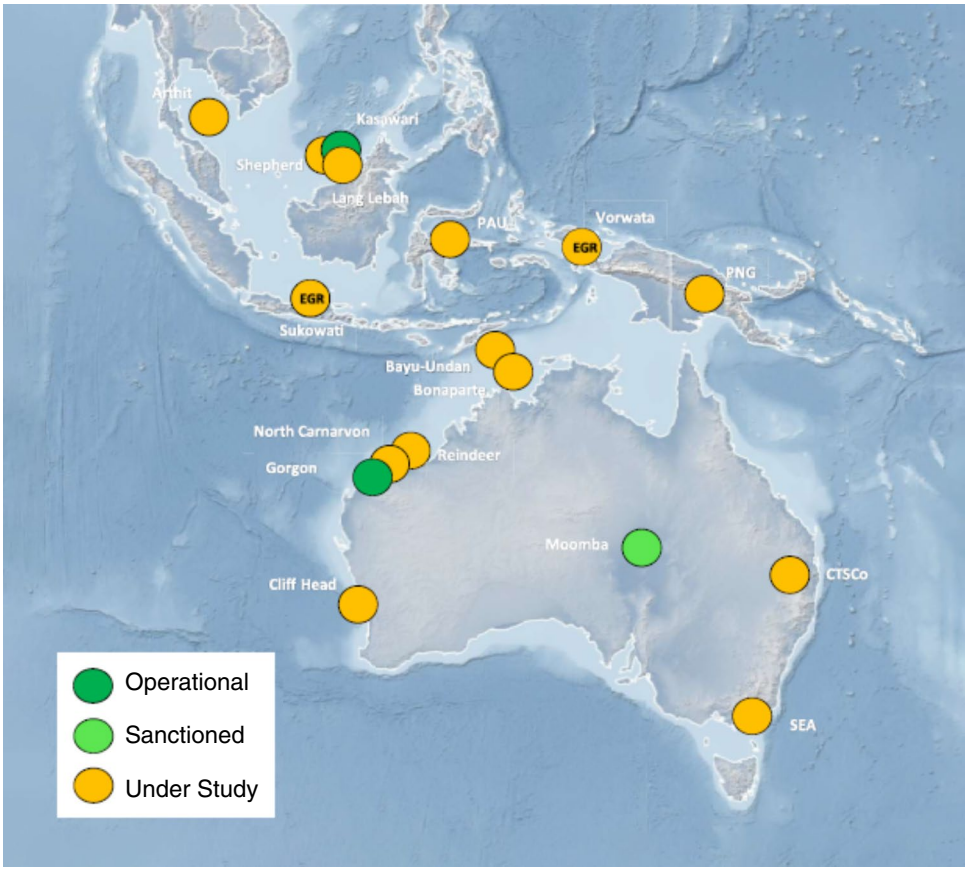


Fig. 1—Location of CCS projects in Southeast Asia and Australia in 2024.

Another primary challenge common to all CCS projects is the difficulty of capturing the carbon from the source gas and the cost of doing so. Capturing CO_2 from natural gas in a gas-processing facility is easier, for instance, than post-combustion carbon capture. Perhaps more importantly, an economic incentive exists to sell the gas, paying for carbon-capture facilities.

The SPE Storage Management System (SRMS) can be used to classify the uncertainty and maturity of CCS projects, with uncertainty being a function of technical parameters and maturity being a function of technical understanding and commercial, legislative, and project support. An SRMS classification could be useful in communicating the status of a project to stakeholders, provided that, throughout the industry, application is consistent and limitations are thoroughly understood.

The publication of the SRMS in 2017, with guidelines following in 2022, initiated a tool to calibrate expectations, uncertainty, and maturity of carbon-storage projects. This identifies three main components—ownership, geological characterization, and project concept—required to estimate storable quantity to be outlined and matured from prospectivity to contingency to capacity classification.

Gorgon CCS Project in Australia

The Gorgon project is approximately 75 km offshore Western Australia. It contains two large gas fields, the Jans and the Gorgon gas fields, from which a facility is fed through subsea pipelines. The facility is a liquefied natural gas (LNG) plant with three LNG trains, with a total export capacity of approximately 16 million tonnes per annum.

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The fields and process facilities are operated by Chevron Australia, with ExxonMobil and Shell as two major joint-venture partners. Gas is produced from the Gorgon and Jans with 14% CO_2 content. The produced gas is captured and separated into an acid-gas-removal unit. The separated CO_2 is then compressed by three compressors and dried before being transported by a 7-km pipeline along Barrow Island, where the LNG plant is located, to one of three drill centers for injection. These centers, with up to nine injectors, are dedicated to injecting CO_2 with rates as high as 150 kg/s, equivalent to 4 million tonnes per annum. Some surveillance wells are dedicated to advanced microseismic and 4D seismic to monitor the CO_2 plume over time.

The CO_2 injection began in August 2019 into a saline reservoir, the Dupuy formation (a regionally extensive clastic formation of late Jurassic deposited in a deepwater-slope depositional setting with an overall thickness of between 200 and 500 m) more than 7,000 ft below Barrow Island. The formation consisted of stacked fine- to medium-grained sands with the Lower and Upper Massive sands targeted for injection. The formation also has excellent extensive and continuous regional seals, making it suitable for CO_2 containment. It is estimated that, over the duration of the Gorgon's 40 years of production life, more than 2 Tcf, or 120 million tonnes, of CO_2, will be injected into the Dupuy formation.

Water is extracted by four water producers and reinjected into a shallower aquifer in the Barrow group. The water withdrawal rate and volume must be balanced such that the injected CO_2 is not produced and is contained in the Dupuy formation.

In general, storage in aquifers is an order of magnitude more complex than in depleted fields. Potential storage volumes in aquifers are significantly greater, however, and depleted reservoirs may be better used for storage of gas that must be reproduced, such as natural gas or hydrogen.

The mechanism of CO_2 injection into aquifers forms a high-saturation, mobile plume of CO_2. The plume will rise under buoyancy forces until it reaches a seal. If it remains above the residual gas saturation, it migrates updip under the seal until it finds a breach in the seal. As long as CO_2 is injected, the mobile plume and migration continues. Once the injection stops, the plume continues to migrate, and CO_2 concentrations disperse until the CO_2 reaches its residual trapped saturation. At this point, the CO_2 becomes immobile or trapped and can no longer move.

The key question is where, and how far, the CO_2 migrates before becoming trapped. Some CO_2 dissolves into the aquifer brine; this continues as the CO_2 contacts fresh brine. In the longer term, the residual or trapped CO_2 saturation reduces as more CO_2 dissolves into the brine. Over thousands of years, CO_2 may mineralize as a solid carbonate.

Gorgon has not shown any signs of injectivity issues, but the water extracted from the Dupuy formation contained far more solid particulates and traced hydrocarbons than expected. This caused considerable impairments in the water-injection wells and a reduction in the injectivity index. Consequently, water extraction had to be reduced, and the pressure of the Dupuy formation could not be reduced by the required amount, which, in turn, limited CO_2-injection capacity.

The water from the Dupuy formation also showed higher pressure, with some clay reactivity. These factors, combined with a slightly lower-than-expected connected pore volume of the formation, created limitations in pressure management and water-handling capabilities. Consequently, CO_2 injection had to be restricted; currently, 1.6 tonnes per year can be sustained.

One of the key lessons from the issues experienced by the project could be to conduct a thorough study of formation damage and fines migration caused by the injection of saline water into a reservoir and to design well completions to take the fines into account.

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Because the behavior, plume movement, and entrapment of the CO_2 in the reservoir is as expected and an efficient surveillance and monitoring process is in place (microseismic and 4D seismic), actions have been focused on pressure management and injectivity of wells into the Barrow formation. At the time of writing, the pressure-management system is functioning reasonably but does not have the ability to reach the desired and planned injection rates. One option is to invest in additional water-handling capacity, even though this measure faces its own challenges from regulatory and environmental approvals.

A more-practical option is to increase the number of water-injection wells from two to four, with horizontal trajectories and gravel-pack completion designs. Also, the existing wells could be sidetracked with sand control and their exposure to the reservoir sandface increased. Chevron is also planning for an additional drilling center and a possible increase in well stocks if needed. In hindsight, Chevron indicated that, if it had begun water extraction at the same time as CO_2 injection and had not waited until the pressure in the Dupuy reservoir increased and necessitated water extraction, then the water-treatment issues could have been discovered earlier and addressed.

Overall, the main reservoir is behaving as expected (albeit with slower injection than originally planned) and a very good match between modeled and actual CO_2-plume movement in the reservoir has been observed. Thus, a high level of confidence in future performance exists once the water-injection problems have been resolved. **JPT**

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